

HARSH RAJ

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EDUCATION

Northeastern University, Boston, MA July 2026 (expected)
Master of Science in Computer Science GPA: 3.6/4.0

Delhi Technological University, Delhi, India August 2023
Bachelor of Technology in Engineering Physics GPA: 3.4/4.0

PROFESSIONAL EXPERIENCE

Research Assistant, Forschungszentrum Jülich July 2025 – Present

- Contribute to EU Open Foundational Model projects ([MINERVA](#), [OpenEuroLLM](#), [ELLIOT](#)) on JUPITER, the EU's largest supercomputer
- Lead ablation studies on dataset mixtures and scaling laws on LLM training data at the petabyte scale
- Optimized multi-node data processing pipelines using Ray, achieving **5x** speedup in deduplication, filtering and map reduction operations

Co-Founder, Ontocord AI (Open Source) Jan 2024 – Present

- Co-founded open-source initiative focused on creating high-quality permissive datasets for LLMs
- Developed data processing pipelines with strategies for mixing, formatting, upsampling, interleaving, and quality validation
- Published [MixtureVitae-300BT](#), the state-of-the-art permissive pretraining dataset to date, achieving 10% performance improvement over existing permissive datasets while matching non-permissive alternatives. Work under review at ICLR 2026.

Applied Scientist, Intern, Vijil AI Apr 2024 – Nov 2024

- Contributed to the [vijil-evaluation](#) platform for red-teaming of enterprise AI agent workflows. Integrated various benchmark datasets for vulnerability assessment
- Developed the vijil-fuzzer framework for adversarial text mutation for red-teaming. It achieved a 80% success rate against GPT-4o at launch

Machine Learning Engineer, Yield Inc Sept 2022 – Mar 2024

- Built synthetic data generation and finetuning infrastructure for LLM agents - [synchaev](#) and [mandrill](#)
- Co-developed [Cacti](#), a Web3 transaction chatbot that simplifies blockchain interactions through natural language

PUBLICATIONS

- Improving Consistency in Large Language Models through Chain of Guidance **1st Author** TMLR 2025
- Defences against Reverse Preference Attacks **3rd Author** AAAI 2025
- Transfer of Adversarial Robustness from Pretraining to Downstream **2nd Author** NeurIPS 2023
- Measuring Reliability of Large Language Models through Semantic Consistency **1st Author** NeurIPS 2022
Best Paper Award

PROJECTS

terminal-bench (600+ GitHub ★) Language agent sandbox evaluation framework (**Co-author & Maintainer**)
inspect-ai (1300+ GitHub ★) Industry-standard LLM evaluation suite (**Contributor**)

AWARDS AND ACHIEVEMENTS

- [SWISS AI](#) Grant Recipient – Awarded 200k GPU hours and CHF 61,000 in funding 2025
- [Vision and Language Navigation](#): Secured 3rd position on the R2R benchmark 2024
- [Rank 7](#) – Codeforces Raif ML Round 1 Competition 2021